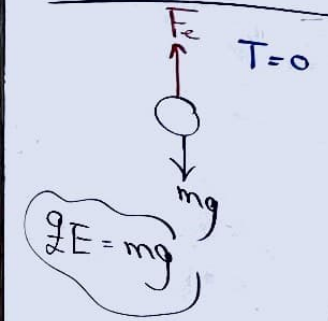
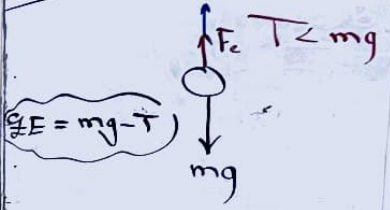
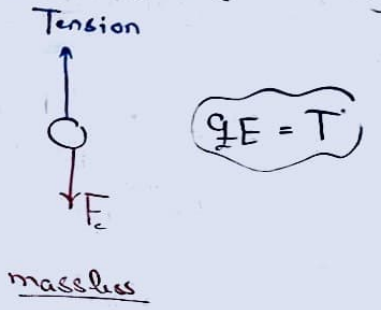
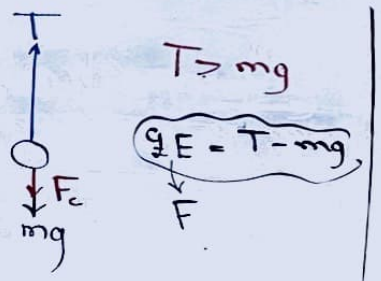


يسمى الجسم متحركاً



suspended charged object (stationary)



moving in E

$$qE = ma \quad a = \frac{qE}{m}$$

Directions

- $+q \rightarrow F \text{ and } a \text{ in the same as } E$
- $-q \rightarrow F \text{ and } a \text{ is opposite to } E$

Example 1.10

$$m = 0.6 \text{ g} = 0.6 \times 10^{-3} \text{ kg} \rightarrow mg = 0.6 \times 10^{-3} \times 9.8 = 5.9 \times 10^{-3} \text{ N}$$

$$E = 300 \text{ N (down)}$$



$$\textcircled{a} \quad q = +8 \text{ nC} = +8 \times 10^{-6} \text{ C} \rightarrow qE = 8 \times 10^{-6} \times 300 = 2.4 \times 10^{-3} \text{ N}$$

$$T = mg + qE = 5.9 \times 10^{-3} + 2.4 \times 10^{-3} = 8.3 \times 10^{-3} \text{ N} \rightarrow 8.3 \text{ mN}$$



$$\textcircled{b} \quad q = -8 \text{ nC} = -8 \times 10^{-6} \text{ C}$$

$$T = mg - qE = 5.9 \times 10^{-3} - 2.4 \times 10^{-3} = 3.5 \times 10^{-3} \text{ N} \rightarrow 3.5 \text{ mN}$$

Q 14

$$m = 0.2 \text{ kg} \rightarrow mg = 0.2 \times 10 = 2 \text{ N} = 1.9 \times 10^3 \text{ N}$$

$$E = 3 \text{ kN} = 3000 \text{ N} \text{ (up)}$$



↑ E (a) T = 0

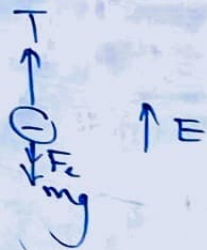
$$\sum F = mg \rightarrow \frac{q}{E} = \frac{mg}{E} = \frac{1.9 \times 10^3}{3000} = 0.63 \times 10^{-3} \rightarrow \frac{q}{E} = + 0.63 \times 10^{-3}$$

(b) T = 4 mN = 4 \times 10^{-3} \text{ N}



$$\sum F = T - mg \rightarrow \frac{q}{E} = \frac{T - mg}{E} = \frac{4 \times 10^{-3} - 1.9 \times 10^3}{3000} = 0.58 \times 10^{-3} \rightarrow \frac{q}{E} = - 0.58 \times 10^{-3}$$

A 0.05 kg charged object is stationary in electric field of 3×10^6 upward if a tension is 505 mN. Find the charge of the object.



$$mg \rightarrow 0.05 \times 9.8 = 0.49 \text{ N}$$

$$T = 505 \text{ mN} = 0.505 \text{ N}$$

$$qE = T - mg \rightarrow \frac{q}{1} = \frac{T - mg}{E} = \frac{0.505 - 0.49}{3000} = 5 \mu\text{C}$$

$$\frac{q}{1} = -5 \mu\text{C}$$